```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
clc;
clear;
close all;
%% Message Signal Parameters
Am = 2; % Message amplitude (V)
fm = 500; % Message frequency (Hz)
%% Carrier Signal Parameters
Ac = 1; % Carrier amplitude (V)
fc = 10000; % Carrier frequency (Hz)
%% FM Parameters
kf = 250; % Frequency sensitivity (Hz/V)
fs = 100000; % Sampling frequency (Hz)
T = 0.01; % Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
delta_f = kf*Am;
%% Modulation Index
beta = delta_f/fm;
%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);
```

```
Carrier Frequency (fc) = 10000 Hz
```

```
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
```

```
Message Frequency (fm) = 500 Hz
```

```
fprintf('Message Amplitude (Am) = %.0f V\n',Am);
```

```
Message Amplitude (Am) = 2 V
```

```
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n',kf);
```

```
Frequency Sensitivity (kf) = 250 Hz/V
```

```
fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);
```

```
Frequency Deviation (Delta f) = 500.00 Hz
```

```
fprintf('Modulation Index (Beta) = %.2f\n',beta);
```

```
Modulation Index (Beta) = 1.00
```

```
%% Verification
beta_verify = delta_f/fm;
fprintf('Verified Modulation Index = %.2f\n',beta_verify);
```

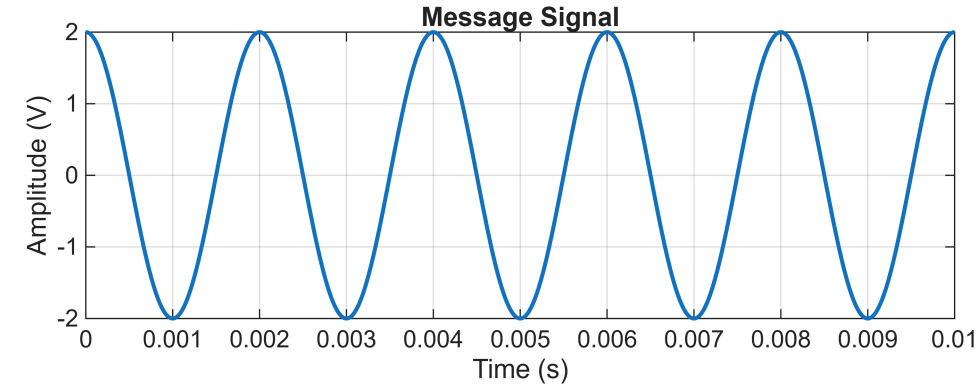
```
Verified Modulation Index = 1.00
```

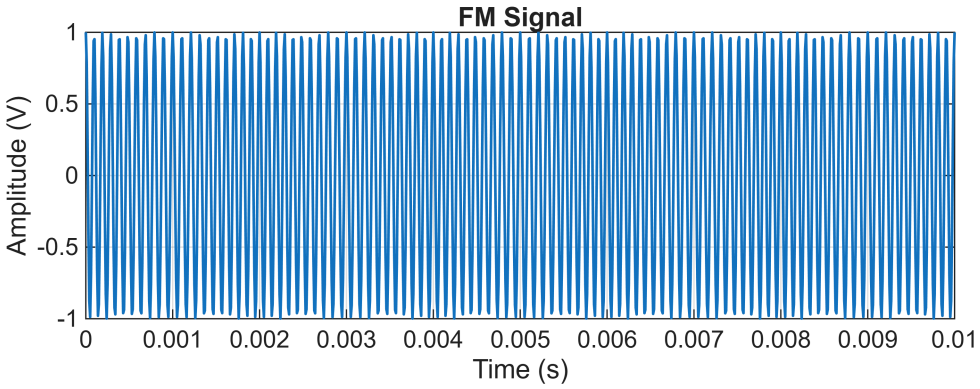
```
%% Verification Check


tolerance = 1e-6;
if abs(beta - beta_verify) < tolerance
    fprintf('\nVerification Successful\n');
else
    fprintf('\nVerification Failed\n');
end
```

```
Verification Successful
```

```
%% Plotting
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
```


Message Signal
Amplitude (V)
2
1
0
-1
-2
0
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
0.009
0.01
Time (s)


FM Signal
Amplitude (V)
1
0.5
0
-0.5
-1
0
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
0.009
0.01
Time (s)